

Listening to Southern Maine: A Data-Driven Look at Community Challenges and Civic Engagement

Hangliang Ren, Muhammad Marzouk Baig

Northeastern University × United Way of Southern Maine

April 2026

Abstract

We analyzed community survey responses collected by the United Way of Southern Maine (UWSM) across Cumberland and York Counties. Using descriptive statistics, chi-square testing, logistic regression, decision trees, principal component analysis, and K-means clustering, we asked three questions: what do residents say their biggest challenges are, which challenges differ by ALICE status, and who feels unheard. The results show housing cost as the dominant concern across every subgroup (82 % of respondents), differences in material-hardship challenges between Below-ALICE and Above-ALICE households, and a finding that Below-ALICE respondents report feeling unheard at a higher rate than their Above-ALICE neighbors. We use these results to inform four concrete recommendations for UWSM’s programming and future data collection.

1 Introduction

The United Way of Southern Maine (UWSM) serves Cumberland and York Counties through direct services, grantmaking, and community advocacy. In late 2025 and early 2026, UWSM fielded a community survey across three instruments of different length to capture residents’ own accounts of the challenges they face, whether they feel represented in community decisions, and whether they are willing to help solve local problems. This report analyzes the resulting Southern Maine sample of 1,702 responses using descriptive statistics, chi-square tests, logistic regression, decision trees, principal component analysis, K-means clustering, and a zero-shot classification pipeline for the two free-text fields. The goal is to answer three questions that guide UWSM’s programming and outreach: what residents identify as their biggest challenges, which challenges differ between households above and below the ALICE threshold, and who feels unheard and is willing to get involved. Section 5 develops the findings and Section 7 translates them into recommendations for UWSM.

2 Background

ALICE (Asset Limited, Income Constrained, Employed) describes households that earn above the federal poverty line but not enough to cover the cost of basic necessities in their region. Federal poverty thresholds are set nationally, have not been substantively revised since the 1960s, and do not track regional costs of housing, child care, healthcare, or transportation, so a household can clear the federal line while still lacking the income to afford a basic monthly budget locally. Because full household-budget data is impractical to collect in a community survey, short instruments use a proxy item; the UWSM survey uses whether the respondent could cover an unexpected \$400 expense, with *Yes* coded as Above ALICE and *No* or *Maybe, with difficulty* coded as Below ALICE. The framework matters because ALICE households are often ineligible for the means-tested programs that serve the lowest-income population yet face the same cost pressures.

The 2026 UWSM community survey provides a direct, first-person snapshot of self-reported needs, financial strain, and community concerns among survey respondents. The dataset includes multiple survey types collected in different settings and with different levels of detail. In the presentation materials, these instruments are discussed as three broad groupings: a Full Survey, abbreviated forms used in shorter collection contexts, and an ALICE Voices instrument (**Community Impact**) that includes more narrative detail. Because item coverage differs across survey types, the analyses below use variable-specific subsets rather than assuming a single identical questionnaire across all responses.

3 Methods and Analysis

3.1 Data source and survey instruments

The data are a single Excel workbook (**UWSM Community Data 4-1-2026.xlsx**) containing an assignment description, a respondent-level data table, and a codebook documenting question text, response options, and canonical recodings. The **Data** sheet contains 1,889 respondent-level records and a **Survey Type** field identifying eight collection channels: **Full Survey**, **3Q**, **Convo CheckIn**, **Community Impact**, **CASH**, **United We Can**, **GiveAtWork**, and **QHC**. Because item coverage differs across instruments, analyses are conducted on variable-specific non-missing subsets rather than assuming a single common questionnaire.

3.2 Sample demographics

Coverage varies by variable and by survey type. Geography and ALICE status are observed for most of the analytic sample, while age, race and ethnicity, relationship to UWSM, feel-heard, and engagement items are available only on subsets of instruments. Per-variable counts are reported in Table 1 and per-field missingness in Table 3; findings about age, race, and civic engagement are therefore interpreted with respect to the respondents who were asked each item rather than the full sample. Employment-status and employment-type columns appear in the raw schema but are sparse and are not used in any inferential analysis.

3.3 Data formatting and survey types

The cleaned dataset retains one row per respondent, indexed by **Unique ID**, with harmonized canonical fields appended to the raw columns. For single-response items, fields not asked on a respondent’s instrument remain missing. The biggest-challenge item, when present, consists of a multi-select checkbox list of 14 predefined options and an optional free-text write-in. The codebook defines 10 additional canonical categories for the write-ins (Affordable Housing Availability, Language Access, Domestic Violence, Rising Taxes, Political Division, Immigration Enforcement, Welfare Dependence, Lack of in-Home Support, Gas Costs, and Rent Increases), plus an *Other* bucket for responses that do not map to any canonical category. Checkbox selections were one-hot encoded directly; free-text responses were classified as described in Section 3.5 and merged onto the same schema, yielding 24 challenge indicators used in the challenge-based models. The feel-heard, want-involvement, and engagement-type items were recoded using the processed labels in the shared analysis scripts; the engagement-type field is a multi-select checkbox that was one-hot encoded across six categories (Volunteer, Speak Up, Join Group, Share Story, Donate, and Connect).

3.4 Data cleaning

Geographic filter. The raw file contains 1,889 responses collected across Maine. In the shared processing pipeline, the Southern Maine analytic sample is defined from the cleaned direct-selection location variable by excluding rows explicitly coded as *Outside Southern Maine*. This retains respondents coded as Cumberland, York, or Cumberland & York and matches the attached processed dataset used in the reported analyses.

ALICE derivation. ALICE status was derived from the \$400-emergency item following the codebook: *Yes* responses are coded as Above ALICE, and *No* and *Maybe, with difficulty* are both coded as Below ALICE. Respondents who did not answer the proxy item are excluded from analyses conditioned on ALICE status and retained in analyses that do not require it.

Feel-heard coding. In the shared processing script, the feel-heard item is collapsed to a binary variable with *Yes* coded as “Heard” and *No* or *Maybe* coded as “Unheard.” All reported feel-heard analyses follow this operational coding. This operational coding differs from the codebook recoding and is stated here to match the attached analysis pipeline.

Missing-data treatment. Missingness was handled on an analysis-specific basis. For single-response variables, analyses use complete cases on the variables entering that analysis and no imputation was performed. For the derived challenge indicators, the preprocessing pipeline converts missing or empty challenge strings to all-zero indicator rows after one-hot encoding.

ZIP-to-county mapping. The ZIP code field was retained in the dataset but not used as the primary geography variable in any analysis. The direct-selection location item was treated as authoritative, and ZIP-to-county resolution was performed only as a spot-check on the direct selection.

3.5 Open-text classification pipeline

Two survey fields contained free-text responses requiring classification before analysis: the biggest-challenge write-ins and the community identity field.

Biggest-challenge write-ins. Write-in responses were classified into the canonical challenge vocabulary using zero-shot classification with `facebook/bart-large-mnli`. Each response was scored against the full category set and assigned the highest-scoring label above a confidence threshold of 0.30; responses below the threshold were coded as Other. The resulting labels were one-hot encoded alongside the checkbox responses to produce a unified 24-column binary feature matrix.

Community write-in field. Responses were processed in two stages. First, a named-entity recognition model (`spacy en_core_web_sm`), supplemented with a manually constructed dictionary of Southern Maine town names, detected geographic references and classified them as Town of Residence. Second, remaining responses were passed to `facebook/bart-large-mnli` for zero-shot classification against eleven community categories at a confidence threshold of 0.25. The pipeline achieved an overall classification rate of 89.5%.

3.6 Statistical methods

All statistical computations used Python with `pandas`, `numpy`, `scipy.stats`, and `scikit-learn`. Random seeds and cross-validation protocols are reported per method below.

Chi-square tests of independence. Pairwise tests were computed with `chi2_contingency`, and expected-cell-count conditions were verified for every reported test. We report Cramér’s V alongside the p -value as an effect-size measure, interpreted with the conventional thresholds $V < 0.1$ weak, $0.1 \leq V < 0.3$ moderate, and $V \geq 0.3$ strong.

Logistic regression. A binary logistic regression predicts Below-ALICE status from the 24 binary challenge indicators using `sklearn.linear_model.LogisticRegression` with `class_weight = "balanced"` to correct for the 30/70 class imbalance. Performance is evaluated using stratified 5-fold cross-validation (`StratifiedKFold`, `shuffle = True`, `random_state = 42`), with accuracy, precision, recall, and F_1 reported as mean and standard deviation across folds. Odds ratios are taken from a single final fit on the full ALICE-valid sample.

Decision tree. A decision tree classifier is fit on the same feature set with `max_depth = 4` and `min_samples_leaf = 10` to constrain complexity and produce interpretable splits; performance is evaluated with the same stratified 5-fold protocol. Feature importances are the normalized mean decrease in Gini impurity from a single final fit, normalized to sum to 1 across the 24 features.

Principal component analysis. PCA is applied to the 24 binary challenge indicators after standardization with `StandardScaler`. Two components are retained, and component loadings and explained-variance ratios are reported in the Results.

K-means clustering. Respondents with at least one cited challenge are segmented with the KMeans estimator from `scikit-learn` on the same 24-column feature matrix, with `n_clusters = 4`, `n_init = 10`, and `random_state = 42`. The choice of k was made after inspecting the inertia curve for $k \in \{2, \dots, 8\}$; seed sensitivity of this choice is flagged in Section 6.

Significance thresholds. Statistical significance is indicated with $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.001$ (***). Effect sizes are reported alongside p -values because the sample is large enough that small, substantively uninteresting differences can reach statistical significance.

4 Results

4.1 Sample at a glance

After filtering out respondents who neither live nor work in Southern Maine, the analytic sample contains 1,702 respondents. Table 1 summarizes the headline counts used throughout the rest of this section.

Table 1: Sample at a glance. Percentages are of the valid-response base for each metric, which varies because not every survey type asked every question.

Metric	Count	% of base
Total respondents (analytic sample)	1,702	—
Cumberland County only	1,114	65 %
York County only	514	30 %
Both counties	73	4 %
Below ALICE threshold	509	30 %
Above ALICE threshold	1,177	70 %
Feel their voice is unheard	315	37 %
Want to get involved	99	19 %

Three quick observations frame the rest of the section. About one in three respondents is Below ALICE, which is already higher than the statewide ALICE rate in most Maine ALICE reports. Roughly one in three respondents who answered the question feel unheard, and about one in five respondents who were asked the engagement question say they want to get involved. We return to this disconnect between dissatisfaction and willingness to act throughout the section.

4.2 Survey coverage matrix

Three survey instruments were merged into one dataset. They share a small common core (ALICE status, location, biggest challenges) but diverge after that. Table 2 shows which questions each

survey actually asked. Every subgroup analysis below is restricted to the survey types that asked the relevant question; we never impute coverage across types.

Table 2: Which questions were asked in each survey type. A check indicates the question was on the instrument; a dash means it was not asked.

	Abbreviated	Full	ALICE Voices
	<i>n=985</i>	<i>n=639</i>	<i>n=78</i>
ALICE status	✓	✓	✓
Location (county)	✓	✓	✓
Biggest challenges	✓	✓	✓
Feel heard	—	✓	✓
Age / Race	—	✓	✓
UWSM relationship	—	✓	✓
Want involvement	—	✓	✓
Community (unheard)	—	✓	✓
Hardest expenses	—	—	✓
Employment	—	—	✓
Roadblocks	—	—	✓
Helpful supports	—	—	✓

The Abbreviated instrument (58% of responses) carries only three items, which is the source of the large demographic missingness we document next.

4.3 Data completeness

Missingness is highly uneven across fields. Table 3 sorts fields by the percentage missing. The pattern is consistent with Table 2: anything asked only on the Full or ALICE Voices instruments is missing for the 58% of respondents who took the Abbreviated version, which pushes baseline missingness for those fields above 50 % before any nonresponse is counted. The 0.9 % missingness on the emergency-\$400 item confirms that when a question was asked, respondents answered it.

Table 3: Missingness across key fields in the analytic sample ($n = 1,702$).

Field	Missing
Community write-in	84.1 %
Race / ethnicity	77.2 %
UWSM relationship	70.5 %
Interest in involvement	69.9 %
Age group	58.8 %
Feel heard	49.4 %
Top challenges	19.2 %
\$400 emergency question	0.9 %

4.4 Top community challenges overall

Of the 1,702 respondents, 1,370 selected at least one biggest challenge. Table 4 ranks the fourteen canonical categories by frequency.

Table 4: Top community challenges across all respondents who cited at least one challenge ($n = 1,370$). Respondents could select multiple challenges, so columns sum to more than 100%.

Challenge	Count	% of $n = 1,370$
Housing cost	1,131	82.6 %
Cost of basics	852	62.2 %
Healthcare cost	688	50.2 %
Mental health	590	43.1 %
Affordable child care access	513	37.4 %
Low wages	348	25.4 %
Behavioral health	279	20.4 %
Affordable elder care access	252	18.4 %
Food access	239	17.4 %
Social isolation	199	14.5 %
Transportation	188	13.7 %
Job instability	152	11.1 %
Education access	36	2.6 %
Climate impacts	26	1.9 %

Housing cost stands alone at the top: 82.6 % of respondents flagged it, a full twenty percentage

points above the next item. The top five (housing, basics, healthcare, mental health, child care) are all above one in three, and together describe a population whose biggest problem is affording everyday life. Climate, political, and education-access concerns, by contrast, were cited by under 5 % of respondents.

4.5 Challenges by ALICE status

Figure 1 compares challenge-citation rates between Below-ALICE and Above-ALICE respondents side by side. Below-ALICE citation rates are higher for the three material-hardship items on the right (food access 10 pp, cost of basics 9 pp, transportation 14 pp), while Above-ALICE respondents cite care-access items more often (child care 12 pp, elder care 8 pp) and healthcare cost 10 pp. Housing cost and mental health are effectively tied, meaning housing stress is a cross-class experience in Southern Maine, not a poverty-specific one.

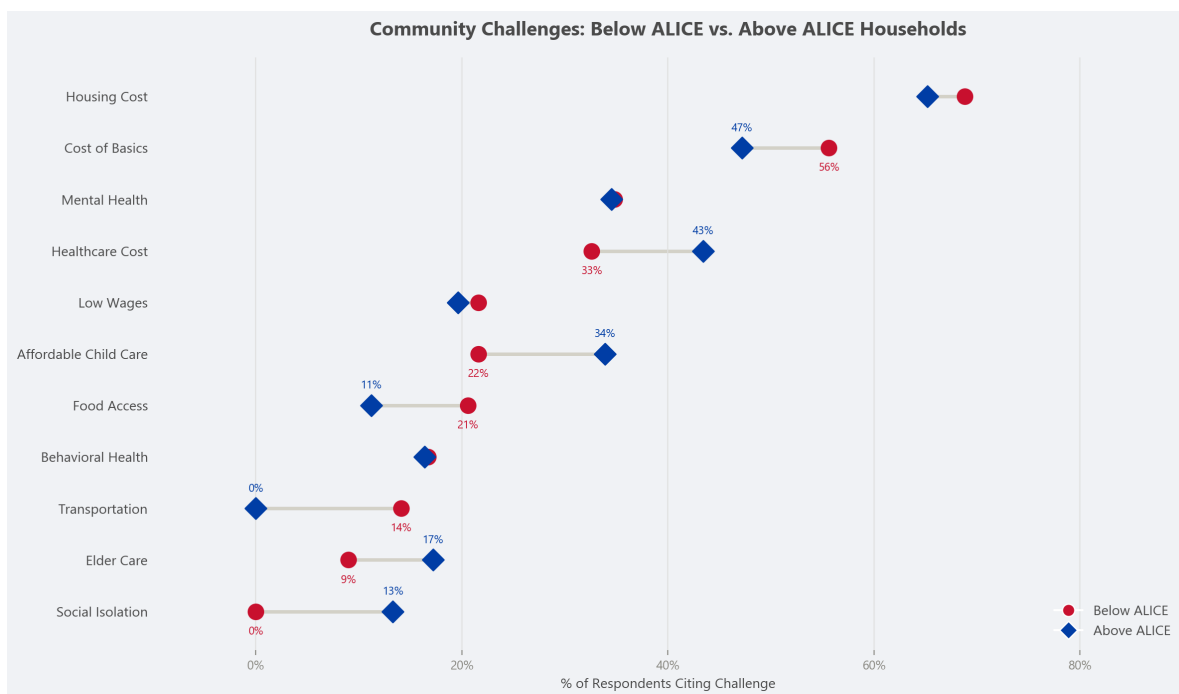


Figure 1: Biggest challenges cited by Below-ALICE versus Above-ALICE respondents. Each row shows the citation rate for one challenge, with the connecting line emphasizing the gap between groups. $n = 1,686$ with valid ALICE status.

Figure 2 reinforces the story. The two word clouds were built from the same free-text biggest-challenge field, split by ALICE status. The central mass of both clouds is nearly identical (housing cost, cost of basics, healthcare cost, mental health), which is what produces the near-zero gap for those items in Figure 1. The difference lives at the periphery: child and elder care access are more visible in the Above-ALICE cloud; education access and transportation are more visible in the Below-ALICE cloud.

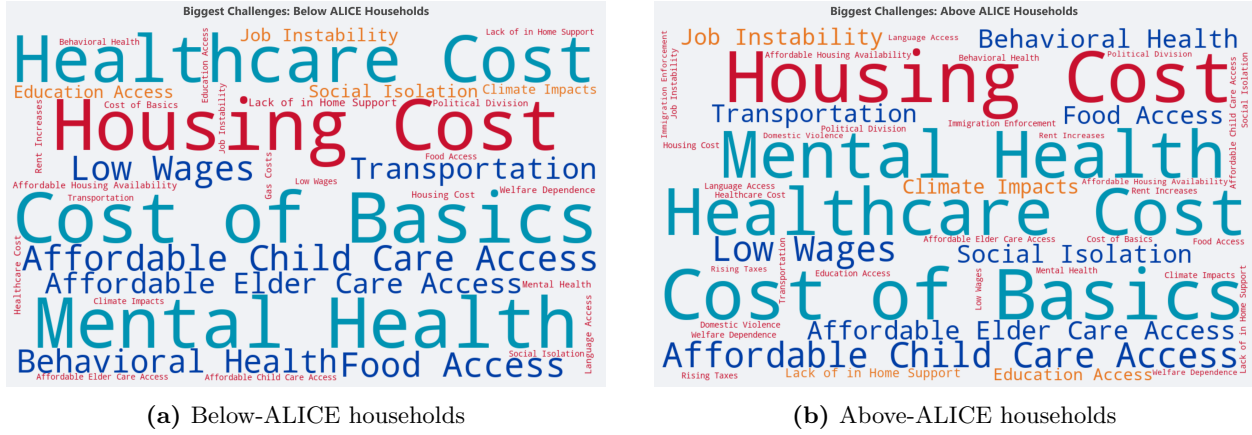


Figure 2: Biggest challenges word clouds by ALICE status. Font size scales with citation frequency. The core concerns are shared; the differences are at the edges.

4.6 Challenges by age

Age effects are striking. Figure 3 shows the seven largest challenges across three age bins. The Under-24 line sits at the bottom for every challenge; every single concern jumps sharply between Under-24 and 25 to 64 respondents. The 25–64 group has the highest rate of child care stress (30 %) and mental health citation (29 %). Respondents 65 and over have the highest rate of housing cost (55 %), healthcare cost (39 %), and elder-care access (21 %), but their mental health citation drops back to 18 %.

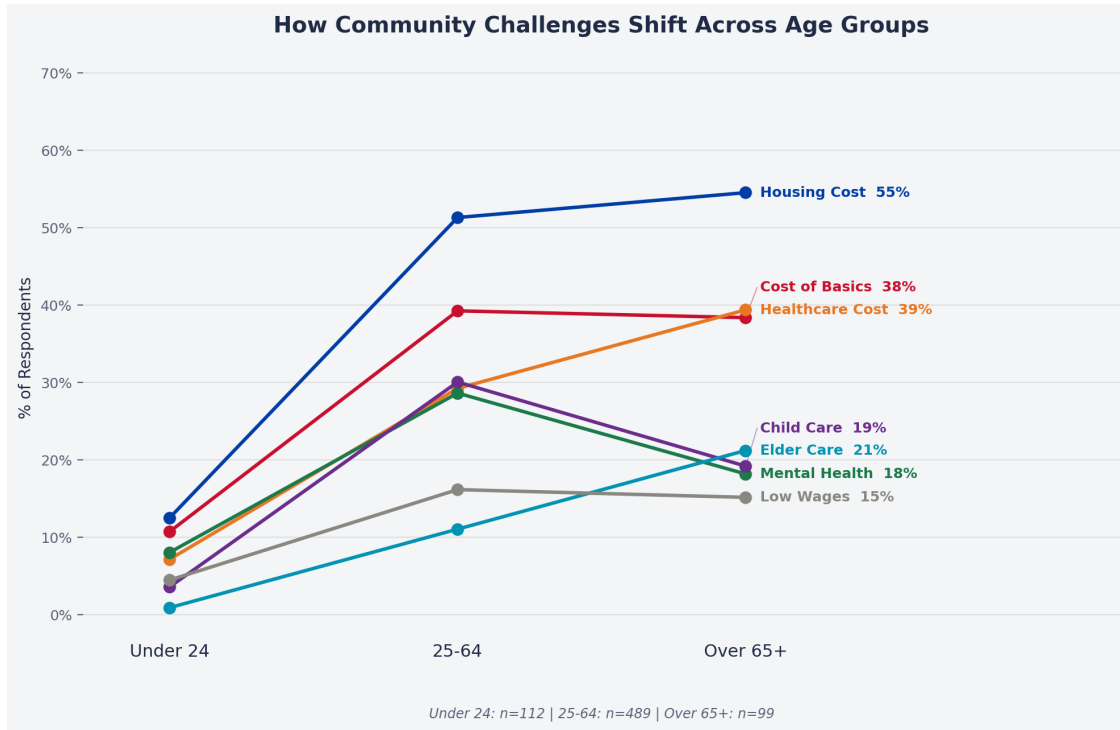


Figure 3: How the seven most common challenges shift across age groups. The y-axis is the percentage of respondents in each age group citing the challenge. Sample sizes by group: Under 24 $n = 112$; 25–64 $n = 489$; Over 65+ $n = 99$.

Two practical takeaways. First, the 25–64 band is the peak-stress window across almost every dimension; programming aimed at working-age residents will reach the group that self-reports the most pressure. Second, the Over-65 group continues to report high material hardship, contradicting the assumption that retirees have stabilized economically.

4.7 Statistical validation: chi-square tests

Table 5 consolidates all three validation panels from the presentation into one set of chi-square tests with Cramér’s V as the effect-size measure. We report only relationships that reached $p < 0.05$; the full 24-row table for challenges $\times$ ALICE is in Appendix A.

Table 5: Chi-square tests of independence across three families of relationships. All cell sizes are large enough to satisfy the expected-count rule. Effect-size interpretation follows conventional thresholds: $V < 0.1$ weak, $0.1 \leq V < 0.3$ moderate, $V \geq 0.3$ strong. Significance codes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Relationship	n	Cramér's V	p	Effect
<i>ALICE status $\times$ individual challenges</i>				
Food access	1,686	0.122***	<0.001	moderate
Affordable child care access	1,686	0.121***	<0.001	moderate
Affordable elder care access	1,686	0.104***	<0.001	moderate
Healthcare cost	1,686	0.100***	<0.001	weak
Cost of basics	1,686	0.075**	0.0019	weak
Social isolation	1,686	0.072**	0.0030	weak
Transportation	1,686	0.065**	0.0079	weak
Education access	1,686	0.054*	0.0273	weak
<i>Feel heard $\times$ demographics</i>				
Race / ethnicity	*	0.214**	<0.01	moderate
ALICE status	*	0.205***	<0.001	moderate
Age group	*	0.072 <i>ns</i>	>0.05	(n.s.)
<i>Want involvement $\times$ demographics</i>				
Race / ethnicity	*	0.224***	<0.001	moderate
ALICE status	*	0.172***	<0.001	moderate
Feel heard	*	0.061 <i>ns</i>	>0.05	(n.s.)
Age group	*	0.082 <i>ns</i>	>0.05	(n.s.)

* Computed on the Full + ALICE Voices subset where the question was asked.

The table makes three things quantitative. First, every challenge-by-ALICE relationship we report has a weak-to-moderate effect size: the differences in Figure 1 are statistically real but not large. Second, both feeling heard and wanting to get involved are significantly tied to race and to ALICE status, but not to age. Third, feeling heard does not predict willingness to get involved: the residents most likely to act are not always the ones who feel listened to.

4.8 Predictive modeling of ALICE status

We next asked whether the challenges a respondent cites can classify them as Below ALICE. We fit two models on the 24 challenge indicators ($n = 1,686$): a class-balanced logistic regression and a decision tree (`max_depth=4`, `min_samples_leaf=10`). Both were evaluated with stratified 5-fold cross-validation.

Table 6: Cross-validated performance of the two classifiers predicting Below-ALICE status. Values are mean $\pm$ standard deviation across five stratified folds.

Model	Accuracy	Precision	Recall	F1
Decision tree (depth 4)	0.582 ± 0.084	0.381 ± 0.053	0.511 ± 0.141	0.421 ± 0.024
Logistic regression	0.556 ± 0.041	0.368 ± 0.024	0.649 ± 0.111	0.466 ± 0.041

Neither model is strong. With a 30/70 class split, a trivial majority-class predictor achieves 70 % accuracy; both our models underperform that baseline. The logistic regression has better recall, which matters if the cost of missing a Below-ALICE respondent is higher than the cost of a false alarm. The practical reading of Table 6 is that the challenges a person reports do not, on their own, cleanly identify whether they are ALICE. This is a substantive finding; we return to it in the Discussion.

Table 7 shows the logistic-regression coefficients as odds ratios: values above 1 increase the odds of being Below ALICE when that challenge is cited, values below 1 decrease them.

Table 7: Logistic regression odds ratios for predicting Below-ALICE status ($n = 1,686$). Challenges are sorted by odds ratio.

Challenge	Odds ratio	Direction
Education access	2.27	increases below ALICE
Gas costs	2.13	increases below ALICE
Food access	1.66	increases below ALICE
Transportation	1.39	increases below ALICE
Cost of basics	1.37	increases below ALICE
Housing cost	1.20	increases below ALICE
Low wages	1.15	increases below ALICE
Mental health	1.04	<i>neutral</i>
Climate impacts	0.90	decreases below ALICE
Language access	0.88	decreases below ALICE
Affordable housing availability	0.71	decreases below ALICE
Healthcare cost	0.64	decreases below ALICE
Affordable elder care access	0.60	decreases below ALICE
Rising taxes	0.58	decreases below ALICE
Social isolation	0.57	decreases below ALICE
Domestic violence	0.56	decreases below ALICE
Affordable child care access	0.55	decreases below ALICE
Immigration enforcement	0.29	decreases below ALICE

Two coefficient clusters are worth naming. The challenges that push a respondent toward Below ALICE are the "survival" items: food, gas, basics, transportation, housing. The challenges that push toward Above ALICE are "infrastructure" items: elder care, child care, healthcare cost. Both groups experience housing pressure, but the basics-of-daily-life concerns discriminate the two groups more than any single financial indicator would.

Table 8 shows the decision-tree view of the same question. It largely agrees with the logistic regression on which signals matter, though the ordering is different because the tree splits greedily on interactions rather than estimating a linear coefficient.

Table 8: Top decision-tree feature importances for predicting Below-ALICE status. Normalized to sum to 1 across all 24 features. Remaining features had importance <0.02 .

Challenge	Importance	Tier
Affordable child care access	0.233	high
Affordable elder care access	0.183	high
Cost of basics	0.148	high
Food access	0.144	high
Social isolation	0.101	high
Healthcare cost	0.097	medium
Behavioral health	0.062	medium
Mental health	0.033	low
Affordable housing availability	0.000	low
Climate impacts	0.000	low

4.9 Dimensionality reduction (PCA)

Principal component analysis on the 24 challenge indicators produced a disappointing result, which is itself informative. Table 9 shows the top five loadings for each of the first two components.

Table 9: Top five loadings for the first two principal components. $n = 1,686$; features standardized. PC1 explains 9.1 % of variance; PC2 explains 7.0 %; together 16.2 %.

PC1 (9.1%)	Loading	PC2 (7.0%)	Loading
Housing cost	0.549	Lack of in-home support	0.650
Cost of basics	0.469	Affordable housing availability	0.474
Healthcare cost	0.347	Domestic violence	0.313
Mental health	0.322	Political division	0.250
Affordable child care access	0.297	Immigration enforcement	0.229

Two components together explain only 16.2 % of the variance in challenge citations, which means challenges do not reduce cleanly to a small number of axes. The qualitative interpretation of the loadings is still useful: PC1 is a "severity of everyday material hardship" axis (high loadings on housing, basics, healthcare, mental health), and PC2 is a "systemic / marginal issue" axis (in-home support, affordable housing, domestic violence, political division). In other words, the first axis captures how hard life is, and the second captures the shape of the hardship. But neither separates Below-ALICE from Above-ALICE respondents by itself, which is consistent with the weak classifier performance in Table 6.

4.10 Respondent segmentation (K-means)

A four-cluster K-means on the challenge indicators produced the segmentation in Table 10. The four segments are distinguished less by severity than by which secondary concern accompanies housing and basics.

Table 10: Four-cluster K-means segmentation of respondents ($n = 1,374$ with at least one challenge cited). "% below ALICE" is the share of the cluster that is Below the ALICE threshold.

Cluster	n	% Below ALICE	Defining challenges (top 3)
1	36	49 %	Education access (100%), housing cost (58%), healthcare cost (50%)
2	215	25 %	Healthcare cost (43%), mental health (42%), cost of basics (33%)
3	179	40 %	Transportation (100%), housing cost (89%), cost of basics (66%)
4	944	29 %	Housing cost (100%), cost of basics (68%), healthcare cost (54%)

Cluster 4 is the dominant "housing-first" segment and contains most of the sample. Cluster 3 (the transportation-plus-housing segment) and Cluster 1 (the education-access segment) have the highest Below-ALICE concentrations and are natural targets for wrap-around services. Cluster 2 is the most stable segment: mostly Above ALICE, centered on healthcare and mental health rather than material hardship.

4.11 Feeling heard and civic engagement

We close Results with three small tables on who feels heard and who is ready to act. Table 11 cross-tabulates feeling heard against ALICE status on the subsample that was asked the question (Full Survey and ALICE Voices). The result is straightforward: the economically less secure Below-ALICE group reports feeling unheard at a higher rate than the Above-ALICE group.

Table 11: Feeling heard by ALICE status (Southern Maine subsample, $n = 848$ with both fields answered). "Unheard" combines the survey's *Yes* and *Maybe* responses; "Heard" corresponds to *No*.

Group	Heard	Unheard	n	% Unheard
Below ALICE	104	117	221	52.9 %
Above ALICE	437	190	627	30.3 %
Total	541	307	848	36.2 %

Table 12 summarizes how the small group of "yes, willing to be involved" respondents want to engage. Preferences are similar across ALICE status for volunteering and speaking up; the gap opens for donating (17 % of Above-ALICE versus 0 % of Below-ALICE, as would be expected) and for sharing one's own story (more common among Below-ALICE respondents).

Table 12: How engagement-ready respondents want to get involved, among those who said yes to involvement (Above ALICE $n = 56$; Below ALICE $n = 43$; total $n = 99$). Respondents could select multiple options.

Engagement type	Above ALICE	Below ALICE
Volunteer	59 %	55 %
Speak up	40 %	40 %
Join a group	36 %	45 %
Share story	19 %	28 %
Donate	17 %	0 %
Connect	5 %	5 %

Finally, Table 13 lists the identity-based communities respondents wrote in when asked which group's voice they feel is unheard. The write-in field was only answered by 242 respondents, so these counts are directional. "Town of residence" dominates (small towns being overlooked), followed by BIPOC, women, older adults, LGBTQ+, and immigrants.

Table 13: Communities respondents said feel unheard, from the open-text community field ($n = 242$ identified a community). "Other" write-ins ($n = 25$) that did not map to a canonical category were excluded.

Community	Count	% of $n = 242$
Town of residence	98	40.5 %
BIPOC	33	13.6 %
Women	22	9.1 %
Older adults (65+)	20	8.3 %
LGBTQ+	19	7.9 %
Immigrant	14	5.8 %
Youth (under 18)	13	5.4 %
Parents	9	3.7 %
Recovery community	8	3.3 %
ALICE	2	0.8 %
Single mothers	2	0.8 %
Hispanic	1	0.4 %
White	1	0.4 %

The dominant write-in is not a demographic group at all. It is a place: the respondent’s own town. That pattern, taken together with Table 11, suggests geography and community identity are as important as economic class for predicting who feels unheard. We develop that reading further in the Discussion.

5 Discussion

5.1 Key findings

The results section surfaces six interlocking findings that together reframe what community hardship and civic disconnection look like in Southern Maine. We discuss each in turn before drawing them together into a unified interpretation.

Housing cost is a cross-class crisis, not a poverty indicator. The most important single number in Table 4 is not a percentage — it is a gap. Housing cost was cited by 82.6 % of respondents, a full twenty percentage points above the next challenge. More revealing is what the chi-square test does *not* find: there is no statistically significant difference in housing-cost citation rates between Below-ALICE and Above-ALICE respondents ($p = 0.168$, $V = 0.034$). The dumbbell chart in Figure 1 makes this visual — the two dots for housing cost sit almost on top of each other.

This result has a direct programming implication. Housing relief framed as an ALICE intervention will miss the majority of people affected. In Southern Maine, housing stress is the ambient condition

of community life, not a marker of the bottom third. Mental health shows the same pattern: cited by 43 % of respondents with no significant group difference ($p = 0.956$). These two items — housing and mental health — define what it means to live in this region regardless of income.

Below-ALICE and Above-ALICE households face structurally different challenges.

Looking past the shared housing and mental-health burden, the two groups diverge sharply. The divergence is not random: it follows a survival-versus-infrastructure logic that holds up across three independent analytic methods, and one that maps naturally onto Maslow’s hierarchy of needs.¹ Below-ALICE households are concentrated at the bottom of the hierarchy — physiological survival and safety — where basic needs such as food, transportation, and everyday expenses account for the most pressing concerns. Above-ALICE households have cleared that threshold and shifted attention upward toward social and esteem-level concerns: child care that enables career participation, elder care that maintains family stability, and healthcare that protects longer-term wellbeing. Critically, the absence of child care and elder care from Below-ALICE challenge profiles does not mean these households have solved those problems — it means they cannot yet afford to name them. When a family is deciding between groceries and rent, the cost of formal child care is not a competing priority; it is an inaccessible aspiration. The hierarchy explains not only what each group reports, but what the lower group *cannot report yet*.

The chi-square tests in Table 5 identify eight challenges with statistically significant differences between groups. Every challenge more common among Below-ALICE respondents is a *material survival* item: food access ($V = 0.122$, $p < 0.001$), transportation ($V = 0.065$, $p < 0.01$), and cost of basics ($V = 0.075$, $p < 0.01$). Every challenge more common among Above-ALICE respondents is an *infrastructure access* item: affordable child care ($V = 0.121$, $p < 0.001$), affordable elder care ($V = 0.104$, $p < 0.001$), and healthcare cost ($V = 0.100$, $p < 0.001$). Social isolation also shows a statistically significant difference ($V = 0.072$, $p = 0.003$) — consistent with the Maslow framing, as social belonging is a higher-order concern that Below-ALICE households have less bandwidth to prioritize.

The logistic regression in Table 7 confirms the same split from a different angle. Survival items carry odds ratios above 1.0 (food access OR = 1.66; transportation OR = 1.39; cost of basics OR = 1.37), while infrastructure items carry odds ratios below 1.0 (child care OR = 0.55; elder care OR = 0.60; healthcare cost OR = 0.64). The decision tree in Table 8 reaches the same conclusion via a third method: child care and elder care are the top two splitting features even though they are more common among Above-ALICE respondents, because their relative *absence* in Below-ALICE profiles is what makes them discriminating.

The most counterintuitive element of this split is that child care and elder care — services one might expect to be concentrated among the most economically vulnerable — are significantly *more* common among Above-ALICE respondents. The explanation lies in what is known as the *benefit cliff*²: a structural gap in the social safety net where households earn just enough to lose eligibility

¹Maslow’s hierarchy of needs overview.

²The benefit cliff refers to the phenomenon where a modest income increase causes a household to lose eligibility

for public subsidies but not enough to absorb market-rate costs. Below-ALICE households often qualify for Head Start, subsidized child care programs, or Medicaid-funded elder care support. Above-ALICE households are priced out of those programs yet cannot comfortably afford market alternatives. They are caught between two systems, and the survey captures exactly that friction. The convergence of chi-square, logistic regression, and the decision tree on the same core set of challenges across three methodologically independent approaches strengthens confidence that this two-tier structure is a genuine feature of the data, not an artifact of any single modeling choice.

Challenges alone cannot classify ALICE status — and that is itself a finding. Table 6 shows that neither classifier performs strongly. With a 30/70 class split, a trivial majority-class predictor achieves 70 % accuracy; both models underperform that baseline on accuracy while only partially recovering on recall (logistic regression recall = 0.649, $F_1 = 0.466$). We interpret this not as a failure of modeling but as a substantive finding about the nature of economic hardship.

It is important to be precise about what the logistic regression was designed to do here. This model was not deployed as a prediction tool — we would not use it to classify new survey respondents as Below or Above ALICE. Rather, it served as a *construct validation exercise*: if the survey genuinely captures the experiences that define economic hardship, then the pattern of challenges a respondent reports should be systematically and predictably linked to their ALICE status. The fact that the model performs better than chance, and that the challenges which predict Below-ALICE status (food, transportation, basics, education access) are precisely the challenges that poverty research identifies as markers of material deprivation,³ is evidence that the survey instrument is measuring something real. Respondents are not answering randomly — their challenge profiles reflect their economic situations. Where the logistic regression estimates independent linear effects, the decision tree captures interaction structure; that both methods identify the same core challenge set is stronger evidence than either model alone would provide.

At the same time, an F_1 of 0.466 tells us that challenge profiles are necessary but not sufficient to identify ALICE status. The survey captures what people *experience*, not the structural conditions that produce those experiences. Several factors that likely account for the remaining variance are absent from the instrument: household savings and debt levels, family size and composition, housing tenure (owner versus renter), health insurance status, and access to informal support networks. A household facing food insecurity but with a working spouse, no debt, and nearby family may clear the \$400-emergency threshold; a household reporting fewer challenges but carrying high fixed costs and no safety net may not. Challenge profiles capture the visible surface of hardship; the underlying determinants are only partially visible through a self-report survey of this type.

Feeling unheard is more common among Below-ALICE respondents. Table 11 shows that Below-ALICE respondents report feeling unheard more often than Above-ALICE respondents (52.9 % vs. 30.3 %). The chi-square test confirms that this difference is statistically significant ($V = 0.205$, $p < 0.001$, moderate effect).

for public assistance programmes, resulting in a net decrease in total resources.

³United For ALICE.

Race and ethnicity add a further dimension. Race/ethnicity is a significant predictor of feeling unheard ($V = 0.214$, $p < 0.01$), with an effect size slightly larger than that of ALICE status ($V = 0.205$). This means structural barriers to civic representation operate along racial lines independently of economic status: a respondent’s identity, not only their income, shapes whether they feel their voice carries weight.

Geographic identity and specific communities are at the center of the unheard story.

Table 13 reveals that the largest single community feeling unheard is not a demographic group at all: it is a place. Town of residence accounts for 40.5 % of all community write-ins, outpacing BIPOC (13.6 %), women (9.1 %), and older adults (8.3 %) combined. Residents are not primarily saying “my racial or economic group is unheard” — they are saying “my town is unheard.”

This finding is worth contextualizing methodologically. The community field was entirely open-ended — respondents could write anything they chose, from a single word to a full sentence. A natural language processing pipeline was required to make sense of the responses; without it, the geographic signal would have remained buried in an undifferentiated “Other” category. The fact that town of residence emerged as the dominant pattern after rigorous classification makes the finding more credible, not less.

The identity-based communities that follow — BIPOC, women, older adults, LGBTQ+, immigrants — represent groups with established literatures on civic underrepresentation. Their appearance here is not surprising, but their rank order relative to town of residence is. For UWSM, this means that hyperlocal civic engagement — connecting residents to their own town’s decision-making processes — may be as high a priority as demographic outreach.

Engagement willingness is low but structured. Only 19 % of respondents expressed willingness to get involved, and the gap between the roughly 37 % who reported feeling unheard and the 19 % who are ready to act is itself a finding: feeling unheard does not translate into civic readiness. The chi-square test confirms that feeling heard does not significantly predict willingness to get involved ($V = 0.061$, $p > 0.05$). What does predict it are ALICE status ($V = 0.172$, $p < 0.001$) and, most strongly, race/ethnicity ($V = 0.224$, $p < 0.001$).

Among the willing minority, preferences are structured by income in a practically meaningful way (Table 12). Volunteering and speaking up are roughly equally preferred across both groups. The divergence is sharp for donating (17 % of Above-ALICE versus 0 % of Below-ALICE) and for sharing one’s story (28 % of Below-ALICE versus 19 % of Above-ALICE). Below-ALICE respondents want to contribute time, voice, and lived experience; they cannot contribute money. Above-ALICE respondents are willing to contribute financially in addition to volunteering. These two profiles call for two distinct engagement strategies rather than a single campaign, and the stronger effect of race over ALICE status suggests those strategies also need to be culturally differentiated.

Synthesis. The six findings connect into a single interpretive frame. Southern Maine faces a layered hardship landscape: a universal top layer of housing and mental health stress that cuts

across all income lines, a survival layer unique to Below-ALICE households concentrated at the base of Maslow’s hierarchy, and an infrastructure layer that traps Above-ALICE households in the benefit cliff between subsidy eligibility and market affordability. The challenge profiles respondents report are real — the logistic regression and decision tree confirm the survey is capturing genuine economic experience — but they are incomplete: structural factors absent from the instrument account for a substantial portion of what makes someone Below ALICE.

Underneath the material story is a civic story that is equally important. Feeling unheard is patterned by economic status, but it is not explained by income alone: Below-ALICE respondents feel more unheard than Above-ALICE respondents, geography drives perceived exclusion as powerfully as demographics, and the residents most willing to act are defined more by their racial and cultural identity than by their income.

The implication for UWSM is not a single program but a portfolio: one that treats housing as the universal crisis it is, bundles survival services for Below-ALICE households, builds bridge programs for Above-ALICE households caught in the benefit cliff, invests in hyperlocal civic infrastructure at the town level, and designs engagement strategies that are simultaneously income-sensitive and culturally specific.

5.2 How UWSM can use these results

The findings above translate into six concrete programming and operational decisions for UWSM.

Treat housing as a universal intervention, not an ALICE programme. Housing cost was cited by 82.6 % of all respondents with no significant difference between income groups. Any housing-relief initiative framed exclusively as support for low-income households will structurally exclude the majority of people affected. UWSM should position housing advocacy and relief as a cross-class community issue and build coalitions that include Above-ALICE stakeholders who have both the financial capacity and the personal motivation to support it.

Bundle survival services for Below-ALICE households. Food access, transportation, and cost-of-basics concerns co-occur in Below-ALICE challenge profiles and are confirmed as the most diagnostic markers of economic hardship by the logistic regression, decision tree, and chi-square analysis. Addressing one in isolation has limited impact when the others remain unresolved. UWSM should consider wrap-around service packages — for example, combining food assistance navigation with transportation support and utility relief — targeted at households at or below the ALICE threshold.

Build benefit-cliff bridge programmes for Above-ALICE households. The infrastructure access challenges most cited by Above-ALICE respondents — child care, elder care, healthcare cost — reflect households that earn too much for public subsidies but too little to absorb market rates. This is the benefit cliff, and it represents an underserved population that UWSM is well-positioned to reach. Sliding-scale child care subsidies, elder care navigation services, and healthcare cost-

sharing programmes targeted at households just above the ALICE threshold would address the gap that neither the public safety net nor the private market currently fills.

Invest in hyperlocal civic engagement at the town level. The finding that town of residence is the single largest unheard community — outpacing every demographic group — means that residents primarily experience civic exclusion as a geographic phenomenon. UWSM should prioritize programming that connects residents to their own town or neighborhood’s decision-making processes: town-hall facilitation, local advisory committees, and community liaison roles in smaller municipalities across Cumberland and York Counties. Demographic outreach remains important, but it should complement rather than substitute for place-based engagement.

Design two parallel engagement tracks. Below-ALICE respondents want to contribute time, voice, and lived experience — not money. Above-ALICE respondents are willing to contribute financially in addition to volunteering. A single campaign asking everyone to get involved will underperform both audiences. UWSM should build two structurally distinct tracks: a volunteer, advocacy, and storytelling track for Below-ALICE participants, and a donor-plus-volunteer track for Above-ALICE participants. Because race/ethnicity is the strongest predictor of engagement willingness ($V = 0.224$), both tracks should be culturally differentiated rather than culturally generic.

Integrate mental health support into financial and housing programmes. Mental health was cited by 43 % of all respondents with no significant difference between ALICE status groups — it is as universal as housing cost. UWSM should explore integrated service models in which mental health screening and referral are embedded into financial coaching sessions, housing navigation appointments, and food pantry visits rather than offered as a separate opt-in programme.

6 Limitations

The findings reported here should be interpreted in light of the following limitations spanning data collection, sample representativeness, and analytical method.

Structural demographic missingness. The most consequential limitation is not random non-response — it is structural. The Abbreviated survey instrument, which accounts for 58 % of all respondents, asked only three questions. Age, race/ethnicity, feeling heard, and willingness to get involved were never presented to those respondents. As a result, demographic missingness exceeds 50 % for most subgroup variables (Table 3), and every demographic subgroup analysis in this report is restricted to the Full Survey and ALICE Voices subsets. Findings about age, race, civic engagement, and community identity should be read as characterizing that subset, not the full sample of 1,702 respondents.

BIPOC subsample too small for reliable inference. The BIPOC subsample contains only $n = 30$ respondents after geographic filtering. Every race/ethnicity finding in this report should

be treated as directional only. The effect sizes are large enough to be substantively interesting and consistent with national research on racial disparities in civic representation, but the sample is too small to rule out sampling variability as an alternative explanation. We retain these findings in the main text because of their theoretical consistency with prior literature, but explicitly caution against drawing policy-level conclusions from them without a better-powered sample.

Self-selection bias. Community surveys are completed by people who choose to complete them. Residents with strong opinions, acute hardship, or existing connections to UWSM are more likely to participate than those who are disengaged, transient, or distrustful of institutions. The sample likely overrepresents the most motivated or most distressed residents and underrepresents those who are hardest to reach.

K-means segmentation is seed-dependent. The four-cluster segmentation reported in Table 10 was produced by a single K-means run with a fixed random seed (`random.state = 42`). K-means is sensitive to the initial placement of cluster centroids, and different seeds can produce meaningfully different cluster assignments. Without running the algorithm across multiple random seeds and computing silhouette scores across a range of k values, the segmentation should be treated as exploratory rather than definitive.

PCA explains limited variance. Principal component analysis on the 24 binary challenge indicators explains only 16.2 % of total variance across the first two components. The two-dimensional reduced space captures a small fraction of the information in respondents' challenge profiles, and any visualization built on that projection should be interpreted with caution.

All findings are associational. No finding in this report supports a causal claim. Chi-square tests and logistic regression establish correlation, not causation. We cannot claim that ALICE status causes civic disengagement, that particular challenges cause economic hardship, or that feeling unheard causes unwillingness to participate.

7 Future Work and Recommendations for the Client

7.1 Recommendations for UWSM

The findings and limitations of this study point to five concrete recommendations for how UWSM should approach future data collection, survey design, and community outreach.

Incentivize survey completion to raise demographic response rates. Race/ethnicity, age, and civic engagement questions were missing for nearly 60 % of respondents because the Abbreviated survey instrument did not ask them. Future rounds should explore structural incentives for completing the Full Survey: shorter question sets, mobile-friendly formats, and community-based administration at events and service sites. The target should be raising demographic response coverage from the current 30 % to at least 70 %.

Collect finer geographic data. County-level geography is too coarse to support neighborhood-level programming decisions. Future surveys should collect zip code or census tract at minimum, and ideally ask respondents to name their town or neighborhood directly. The NLP pipeline developed for this study already detects and classifies town names from free-text responses and could be repurposed to standardize and geocode those responses automatically.

Add structural questions to capture causes, not only symptoms. The current survey identifies what residents experience but not why they experience it. Appending a small number of structural questions to future rounds — current debt load, housing tenure, household size, health insurance status — would move UWSM from diagnosing symptoms to understanding the conditions that produce them, enabling more targeted intervention at the root cause level.

Invest in BIPOC outreach and oversampling. Race/ethnicity is the strongest predictor of both feeling unheard and engagement willingness in the current data, yet the BIPOC subsample of $n = 30$ is too small to draw statistically reliable conclusions. UWSM should establish partnerships with community organizations serving Black, Indigenous, Hispanic, and immigrant populations to actively recruit BIPOC respondents, with a minimum target of $n = 150$ in future rounds.

7.2 Analytic extensions for a future round

Several analytical extensions would strengthen the findings of this study and open new lines of inquiry for future rounds.

Multivariate modeling. Adding demographic covariates — age, race/ethnicity, and county — to the logistic regression would allow for testing whether ALICE status remains a significant predictor of feeling unheard after controlling for demographics, separating the income effect from the identity effect.

Longitudinal tracking. Repeating the survey annually with a consistent instrument would allow UWSM to measure whether challenge profiles and civic sentiment shift over time, and whether interventions are producing measurable change.

Test-retest reliability for the feel-heard item. Administering the feel-heard question twice to the same respondents would establish whether the item is stable over time and whether the binary Heard/Unheard coding faithfully captures the underlying construct.

Oversampling BIPOC respondents. The current BIPOC subsample of $n = 30$ is insufficient for statistically reliable subgroup analysis. Targeted outreach to reach a minimum of $n = 150$ BIPOC respondents would enable properly powered race/ethnicity comparisons in future rounds.

K-means validation. Running the clustering algorithm across multiple random seeds and computing silhouette scores across a range of k values would confirm whether the four-cluster solution is stable or an artifact of a single initialization.

Integration with external datasets. Linking survey responses to publicly available data — American Community Survey estimates, SNAP enrollment rates, and housing price indices by census tract — would allow findings to be triangulated against objective measures of hardship.

A Race comparison (directional only)

Figure 4 shows the challenge-citation gap between White and BIPOC respondents. With BIPOC $n = 30$, the differences are directional only and are not formally tested. We include the figure in the appendix for completeness because collecting a better race sample is a recommendation to the client.

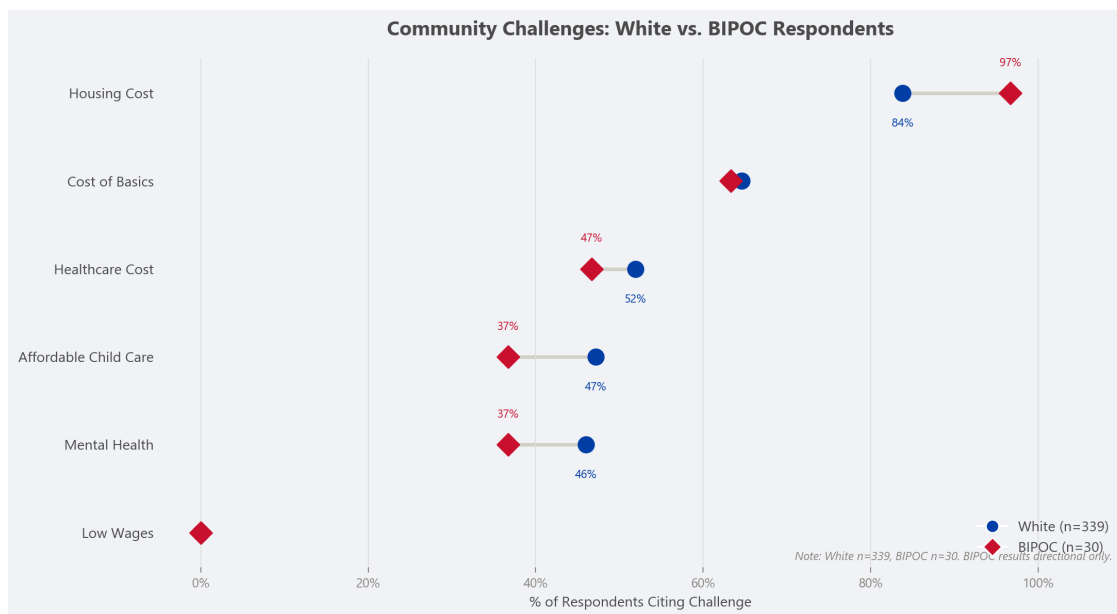


Figure 4: Community challenges cited by White and BIPOC respondents. White $n = 339$, BIPOC $n = 30$. Small BIPOC sample size precludes significance testing.

B Code and data availability

All analysis code, the cleaning pipeline, the processed dataset, and the full figure set are available at: https://github.com/myhott163com/united_way_final.